

Multiobjective Optimization of Dual-Purpose Power Plants and Water Distribution Networks

Ramón González-Bravo,[†] Fabricio Nápoles-Rivera,[†] José María Ponce-Ortega,^{*,†} and Mahmoud M. El-Halwagi^{‡,§}

[†]Chemical Engineering Department, Universidad Michoacana de San Nicolás de Hidalgo, Morelia, Michoacán 58060, México

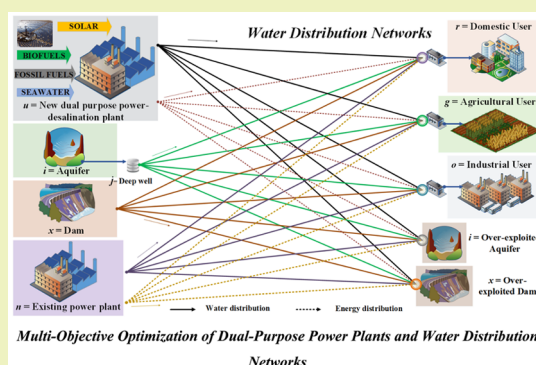
[‡]Chemical Engineering Department, Texas A&M University, College Station, Texas 77843, United States

[§]Department of Chemical and Materials Engineering, Faculty of Engineering, King Abdulaziz University, P.O. Box 80204, Jeddah, 21589, Saudi Arabia

S Supporting Information

ABSTRACT: This paper presents a multiobjective optimization approach for synthesizing water distribution networks involving dual-purpose power plants. The proposed model accounts for environmental, economic, and social objectives by accounting for greenhouse gas emissions, jobs, and net profit. The model considers water and energy demands for domestic, agricultural, and industrial users. Energy is provided through several alternatives including fossil fuels (i.e., natural gas and oil), biofuels (i.e., biomass, biogas, biodiesel, and bioethanol), and solar energy. Water demands are satisfied by fresh water from dams, lakes, rivers, aquifers, and artificial storage tanks. The proposed model is applied to a case study from the Mexican State of Sonora. The results show the viability of the dual-purpose power—water plants, the merits of incorporating solar energy in the system, and the economic, environmental, and social benefits of applying the proposed approach. The optimal solution yields a total annual profit of \$MM 1,545.9, it generates 1.37×10^7 ton CO₂ equiv/y and 19 781 jobs.

KEYWORDS: Desalination, Water distribution networks, Optimization, Power production, Dual-purpose power plants



INTRODUCTION

Water scarcity is one of the main problems around the world. By the year 2030, a global water deficit of 40% is expected due to population growth, migration, urbanization, and industrialization.^{1,2} Furthermore, climate change and variability are likely to have a negative impact on fresh water resources. The problem has been experienced more severely in the driest areas of the planet. To address the water shortage problems, several strategies have been proposed.³ These options include extraction of water from aquifers, interstate transportation of water, and constructions of dams and artificial lakes. Other strategies include the installation of seawater and brackish water desalination plants as well as dual-purpose power plants, where water and electricity are simultaneously generated. Currently, more than 86 million m³ per day of fresh water streams are produced using desalination technologies.^{4–6} The International Desalination Association (IDA) reports that there are around 18 000 desalination plants worldwide with most of the installed desalination plants using reverse osmosis (RO) and multistage flash (MSF) desalination techniques. It is worth noting that most of these plants are coupled with dual-purpose power plants producing simultaneously water and electricity.³ Coproduction of water and power is a common practice, especially in Middle East and North African (MENA) countries, where the lack of fresh water has

prompted the installation of large-scale dual-purpose power plants.^{4,5} The simultaneous production of water and energy has significant benefits in terms of sustainability, Raluy et al. reported that a dual-purpose power plant can reduce the overall environmental impact by 75% compared with thermal desalination technologies.⁷ More opportunities are possible and the technological innovation is essential in developing new energy-efficient technologies to produce water and energy by reducing the cost and the environmental impact.⁸

Fossil fuels play an important role in power plants, where natural gas and coal remain being the main sources of energy to produce water and electricity; however, because the used fossil fuels in power plants are responsible of over 65% of the estimated carbon dioxide (CO₂) emissions,^{9,10} renewable energy can play an important role in desalination and power generation by reducing the CO₂ emissions that contribute to the global warming and fossil resource depletion.¹¹ The use and development of renewable energy technologies may provide new pathways for sustainable water and energy production to meet future demands.^{12–15} Several countries (such as Qatar, Egypt,

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Saudi Arabia, UAE, and Jordan) have developed large solar-powered plants to provide electricity and water.^{16,17} Solar energy and biofuels represent great potential alternatives to provide heat. Nonetheless, the use of biofuels and solar technologies should be subjected to techno-economic assessment.

Recently, some approaches have been reported for the proper use of water in industrial parks¹⁸ and refineries.¹⁹ Some others have been proposed to enhance the efficiency of power production and desalination involving renewable energy. Gutierrez-Arriaga et al. developed a multiobjective optimization procedure for designing integrated steam power plants, where economic and environmental objectives were considered.²⁰ Ianquaniello et al. presented an economic analysis for water desalination (involving MSF and RO) powered by solar energy, this model accounts for energy consumption and greenhouse gas emissions (GHGE), the results showed that the price of water in such systems is close to the price of water produced through conventional fossil-based desalination systems.²¹ Liqreina and Qoaider presented a simulation of a concentrating solar power (CSP) in southern Spain, the results showed that the proposed power plant can work better in dry regions.²² This work was later extended by including an optimization procedure.²³ Fthenakis et al. presented an analysis of RO desalination powered by solar energy (PV), the analysis was performed using small (6550 m³/day) and large (190 000 m³/day) capacities, the results showed that the PV-RO power plants have the potential to replace 19 million m³ of diesel fuel per year along with the corresponding reduction in CO₂ emissions.²⁴ Sankar et al. presented an analysis of integrated solar power and desalination plants, where concentrated solar power plants and biomass can be used, the results showed several savings in internal energy consumption and investment cost.²⁵ Gorjian and Ghobadian presented a comprehensive analysis for the potential installation of a water desalination plant powered by solar energy with reduction in CO₂ emissions due to the integration of solar energy.²⁶ Abdelhady et al. developed an optimization approach for the cost-effective conservation of energy and reduction of GHG emissions, the model was applied to an industrial process coupled with cogeneration systems including fossil fuels and solar energy.²⁷ Palenzuela et al. presented a techno-economic analysis of several seawater desalination systems with concentrated solar power plants applied to the MENA region.^{28,29} Diaf et al. proposed a techno-economic analysis for solar desalination plants based on a multiple tray desalination unit running with solar energy to satisfy small scale water demands.³⁰

In addition to the importance of integrating energy and water in yielding sustainable development strategies, it is also important to address the water distribution issues through a macroscopic approach. Liu et al. introduced a mixed-integer linear programming (MILP) problem for the management, production, distribution and storage of reclaimed and desalinated water accounting for the location and capacity of the new desalination plant, pipelines, pumping stations, wastewater treatment, and reclaimed water plants.³¹ Atilhan et al. developed a mathematical model to design macroscopic water networks, the model accounted for water sources (seawater, underground water and desalination plants) and water demands, which allows identifying the location and capacity of new plants. This work was extended later to include monthly demand fluctuations, water storage and wastewater treatment.^{32,33} Nápoles-Rivera et al. introduced a mathematical programming formulation for designing water distribution networks considering economic and sustainability aspects, this model includes sources (dams,

springs and deep wells) and domestic and agricultural users. Later, this model was extended to include alternative water sources (harvested rainwater), seasonal availability (natural sources), population growth (water demands) and industrial water demands.^{34,35} Alnouri et al. presented an energy management approach involving water desalination and power generation, where a deterministic MILP problem was used to identify the optimal allocation of desalination plants, capacities, technologies and energy sources, this model also took into account the variation of water and electricity demands, the results showed significant cost reductions.³⁶ Recently, González-Bravo et al.³⁷ proposed a single objective optimization model for synthesizing water distribution networks involving power-desalination plants, this model was applied in a water stressed region of Mexico located in the desert of Sonora accounting for the water in lakes, dams, aquifers, and rivers, as well as the water in existing storage tanks. It should be noticed that this approach did not take into account the possibility to integrate multiple fuels as well as the possibility to integrate solar energy to supply the power-desalination plant. These are interesting options, because the region of the desert of Sonora has great solar radiation, and also the use of different fuels may impact positively the sustainability of the system.

While previous research activities have tackled the problems of design dual-purpose power-water desalination plants and water-management networks separately, there is a need to address both problems simultaneously and to introduce the use of renewable energy while considering multiple design objectives. This paper presents a multiobjective optimization model to determine the optimal water distribution network involving dual-purpose power plants powered with renewable energy. The proposed multiobjective optimization model simultaneously addresses economic, environmental, and social objectives.

■ PROBLEM STATEMENT

The problem statement is described as follows: Given a macroscopic water distribution problem with

- water and electricity demands for domestic users which have seasonal variations through the year
- water and electricity demands for industrial users which remain constant through the year
- water and electricity demands for agricultural users which depend on the cropping season.
- industrial and agricultural users which have seasonal variations through the year
- water demands that can be satisfied by the existing dams, aquifers, and rivers of each region (the availability of water depends on the natural recharge, precipitation, and extractions)
- electricity demands that can be satisfied by the existing power plants of the region
- water and energy demands that can be satisfied by the installation of new dual-purpose power plants
- overexploited resources (dams, aquifers, and rivers) that can be recharged by the water produced in dual-purpose power plants
- existing and new storage tanks in the considered regions which are also part of the distribution model
- energy requirements for the dual-purpose power plants that can be satisfied using fossil fuels (i.e., natural gas and oil), biofuels (i.e., biomass, biogas, biodiesel, and bioethanol), and solar energy

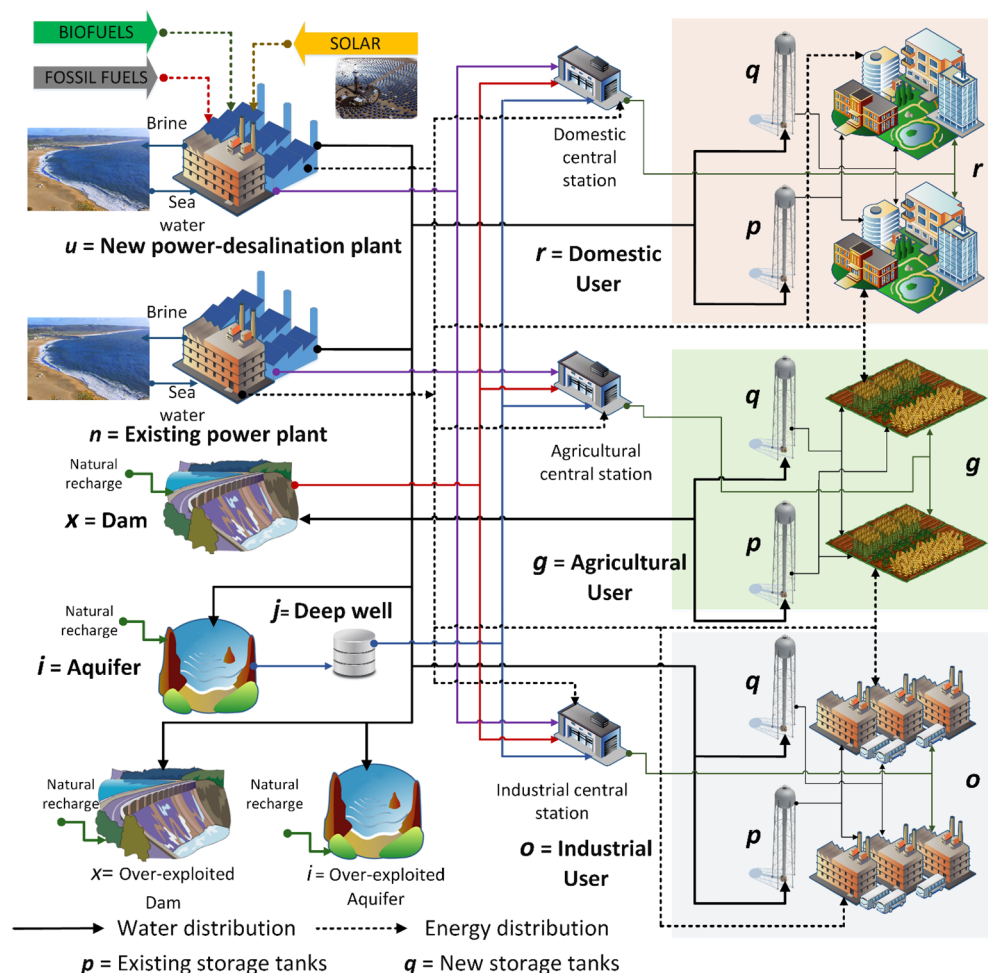


Figure 1. Proposed superstructure for water distribution integrated to dual-purpose power plants.

- use of fossil fuels and biofuels which is subject to the maximum availability in each time period

The main problem consists of finding the optimal water distribution network accounting for economic, environmental, and social objectives. The model seeks to satisfy energy and water demands accounting for the net annual profit, as well as the overall greenhouse gas emissions (GHGE). Figure 1 is a schematic representation of the proposed superstructure embedding the potential configurations of the design alternatives.

MATHEMATICAL MODEL

The water distribution network can be described by tracking mass, energy, and economic aspects. The mathematical model is based on the superstructure shown in Figure 1. The water and electricity demands are tracked for domestic users (r), industrial users (o), and agricultural users (g); the water demands can be satisfied by the water volume content in the existing aquifers of the region (i) by extracting water from the existing deep wells in each region (j) and by the existing water in the dams of the region (x). In this case, the model considers the location of existing storage tanks (p) as well as the possible installation of new storage tanks (q). The electricity can be generated in the existing power plants of the region (n) as well as the possible installation of new dual purpose plants (u). The energy requirements of the dual purpose power plant can be fully satisfied using fossil fuels (f), biofuels (b), and/or solar energy. The model is divided in

time periods (t) (i.e., months). The proposed model includes linear, nonlinear, and logical relationships, which are described as follows.

Model for Water Distribution. The change in the total water volume of an aquifer over a certain time period ($W_{i,t} - W_{i,t-1}$) is equal to the sum of the of water received from existing storage tanks ($s_{p,i,t}^{E,eq}$), new storage tanks ($s_{q,i,t}^{N,eq}$), existing power desalination plants ($B_{n,i,t}^{E,des}$), new power desalination plants ($B_{u,i,t}^{N,des}$), the recharge from agricultural crops ($F_{i,t}^{agr}$), and the natural recharge ($R_{i,t}^{aq}$) minus the water that is sent to deep wells ($a_{i,j,t}^{dw}$):

$$W_{i,t} - W_{i,t-1} = \sum_p s_{p,i,t}^{E,eq} + \sum_q s_{q,i,t}^{N,eq} + \sum_n B_{n,i,t}^{E,des} + \sum_u B_{u,i,t}^{N,des} + F_{i,t}^{agr} + R_{i,t}^{aq} - \sum_j a_{i,j,t}^{dw}, \quad \forall i \in I, \forall t \in T, t \neq 1 \quad (1)$$

The water in deep wells ($a_{i,j,t}^{dw}$) is extracted by the distribution stations including domestic stations ($d_{r,j,t}^{dom}$), industrial stations ($d_{o,j,t}^{ind}$), and agricultural stations ($d_{g,j,t}^{agr}$):

$$\sum_i a_{i,j,t}^{dw} = \sum_r d_{r,j,t}^{dom} + \sum_g d_{g,j,t}^{agr} + \sum_o d_{o,j,t}^{ind}, \quad \forall j \in J, \forall t \in T \quad (2)$$

The total water in each central station (h_{it}) is equal to the sum of the water received from existing power desalination plants ($v_{g,n,t}^E$), new power desalination plants ($v_{g,u,t}^N$), deep wells ($d_{g,j,t}$), and dams ($w_{g,x,t}$).

For domestic stations

$$h_{r,t}^{\text{dom}} = \sum_n v_{r,n,t}^{E,\text{dom}} + \sum_u v_{r,u,t}^{N,\text{dom}} + \sum_j d_{r,j,t}^{\text{dom}} + \sum_x w_{r,x,t}^{\text{dom}}, \quad \forall r \in R, \forall t \in T \quad (3)$$

For industrial stations

$$h_{o,t}^{\text{ind}} = \sum_n v_{o,n,t}^{E,\text{ind}} + \sum_u v_{o,u,t}^{N,\text{ind}} + \sum_j d_{o,j,t}^{\text{ind}} + \sum_x w_{o,x,t}^{\text{ind}}, \quad \forall o \in O, \forall t \in T \quad (4)$$

For agricultural stations

$$h_{g,t}^{\text{agr}} = \sum_n v_{g,n,t}^{E,\text{agr}} + \sum_u v_{g,u,t}^{N,\text{agr}} + \sum_j d_{g,j,t}^{\text{agr}} + \sum_x w_{g,x,t}^{\text{agr}}, \quad \forall g \in G, \forall t \in T \quad (5)$$

The water demands (domdem) can be satisfied by the volume of water in central station ($h_{r,t}$), plus the water in existing storage tanks ($s_{p,r,t}^E$) and new storage tanks ($s_{q,r,t}^N$).

For domestic users

$$\text{domdem}_{r,t} = h_{r,t}^{\text{dom}} + \sum_p s_{p,r,t}^{E,\text{dom}} + \sum_q s_{q,r,t}^{N,\text{dom}}, \quad \forall r \in R, \forall t \in T \quad (6)$$

For industrial users

$$\text{inddem}_{o,t} = h_{o,t}^{\text{ind}} + \sum_p s_{p,o,t}^{E,\text{ind}} + \sum_q s_{q,o,t}^{N,\text{ind}}, \quad \forall o \in O, \forall t \in T \quad (7)$$

For agricultural users

$$\text{agrdom}_{g,t} = h_{g,t}^{\text{agr}} + \sum_p s_{p,g,t}^{E,\text{agr}} + \sum_q s_{q,g,t}^{N,\text{agr}}, \quad \forall g \in G, \forall t \in T \quad (8)$$

A percentage of the used water for agriculture (pca) seeps into the ground, thus it may recharge the aquifer ($F_{i,t}^{\text{agr}}$) as follows:

$$F_{i,t}^{\text{agr}} = \sum_{g(i)} \text{pca} \times \text{agrdom}_{g,i,t} \quad \forall i \in I, \forall t \in T \quad (9)$$

The total water volume in a dam in a certain time period ($M_{x,t} - M_{x,t-1}$) is equal to the sum of the water received from existing power desalination plants ($G_{u,x,t}^{E,\text{rel}}$), new power desalination plants ($G_{u,x,t}^{N,\text{rel}}$), and the water naturally recharged ($R_{x,t}^{\text{dom}}$), minus the water sent to domestic stations ($w_{r,x,t}^{\text{dom}}$), industrial stations ($w_{o,x,t}^{\text{ind}}$), and agricultural stations ($w_{g,x,t}^{\text{agr}}$):

$$M_{x,t} - M_{x,t-1} = \sum_n G_{u,x,t}^{E,\text{rel}} + \sum_u G_{u,x,t}^{N,\text{rel}} + R_{x,t}^{\text{dam}} - \sum_r w_{r,x,t}^{\text{dom}} - \sum_g w_{g,x,t}^{\text{agr}} - \sum_o w_{o,x,t}^{\text{ind}}, \quad \forall x \in X, \forall t \in T, t \neq 1 \quad (10)$$

The total water volume in existing storage tanks in a certain time period ($S_{p,t}^E - S_{p,t-1}^E$) is equal to the sum of the water received from existing power desalination plants ($D_{n,p,t}^{E,\text{Esto}}$) and new power desalination plants ($D_{u,p,t}^{N,\text{Esto}}$), minus the water sent to domestic stations ($s_{p,r,t}^{E,\text{dom}}$), industrial stations ($s_{o,p,t}^{E,\text{ind}}$), agricultural stations ($s_{g,p,t}^{E,\text{agr}}$) and the water sent to aquifers ($s_{p,i,t}^{E,\text{aq}}$):

$$S_{p,t}^E - S_{p,t-1}^E = \sum_n D_{n,p,t}^{E,\text{Esto}} + \sum_u D_{u,p,t}^{N,\text{Esto}} - \sum_r s_{p,r,t}^{E,\text{dom}} - \sum_g s_{g,p,t}^{E,\text{agr}} - \sum_o s_{o,p,t}^{E,\text{ind}} - \sum_i s_{p,i,t}^{E,\text{aq}}, \quad \forall p \in P, \forall t \in T, t \neq 1 \quad (11)$$

The total water volume in new storage tanks in a certain time period ($S_{q,t}^N - S_{q,t-1}^N$) is equal to the sum of the water received from existing power desalination plants ($D_{u,q,t}^{E,Nsto}$) and new power desalination plants ($D_{n,q,t}^{N,Nsto}$), minus the water sent to domestic stations ($s_{q,r,t}^{N,\text{dom}}$), industrial stations ($s_{o,q,t}^{N,\text{ind}}$), agricultural stations ($s_{g,q,t}^{N,\text{agr}}$), and the water sent to aquifers ($s_{q,i,t}^{N,\text{aq}}$):

$$S_{q,t}^N - S_{q,t-1}^N = \sum_n D_{n,q,t}^{E,Nsto} + \sum_u D_{u,q,t}^{N,Nsto} - \sum_r s_{q,r,t}^{N,\text{dom}} - \sum_g s_{g,q,t}^{N,\text{agr}} - \sum_o s_{o,q,t}^{N,\text{ind}} - \sum_i s_{q,i,t}^{N,\text{aq}}, \quad \forall q \in Q, \forall t \in T, t \neq 1 \quad (12)$$

The amount of seawater extracted from the sea by the existing power desalination plants ($\text{SW}_{n,t}^{\text{in},E}$) is equal to the sum of water sent to the existing storage tanks ($D_{n,p,t}^{E,\text{Esto}}$), new storage tanks ($D_{u,p,t}^{E,Nsto}$), domestic stations ($v_{r,n,t}^{E,\text{dom}}$), industrial stations ($v_{o,n,t}^{E,\text{ind}}$), and agricultural stations ($v_{g,n,t}^{E,\text{agr}}$); also the water sent to recharge aquifers ($B_{n,i,t}^{E,\text{des}}$), the water sent to recharge dams ($G_{n,x,t}^{E,\text{rel}}$), and the water sent to the sea as reject ($b_{n,t}^{E,\text{rej}}$):

$$\text{SW}_{n,t}^{\text{in},E}(1 - \beta) = \sum_p D_{n,p,t}^{E,\text{Esto}} + \sum_q D_{u,q,t}^{E,Nsto} + \sum_r v_{r,n,t}^{E,\text{dom}} + \sum_g v_{g,n,t}^{E,\text{agr}} \quad \forall n \in N, \forall t \in T + \sum_o v_{o,n,t}^{E,\text{ind}} + \sum_i B_{n,i,t}^{E,\text{des}} + \sum_x G_{n,x,t}^{E,\text{rel}} + b_{n,t}^{E,\text{rej}} \quad (13)$$

The amount of seawater extracted from the sea by the new power desalination plants ($\text{SW}_{u,t}^{\text{in},N}$) is equal to the sum of water sent to the existing storage tanks ($D_{u,p,t}^{N,Esto}$), new storage tanks ($D_{n,p,t}^{N,Nsto}$), domestic stations ($v_{r,u,t}^{N,\text{dom}}$), industrial stations ($v_{o,u,t}^{N,\text{ind}}$) and agricultural stations ($v_{g,u,t}^{N,\text{agr}}$), also the water sent to recharge aquifers ($B_{u,i,t}^{N,\text{des}}$), the water sent to recharge dams ($G_{u,x,t}^{N,\text{rel}}$) and the water sent to the sea as reject ($b_{u,t}^{N,\text{rej}}$):

$$\text{SW}_{u,t}^{\text{in},N}(1 - \beta) = \sum_p D_{u,p,t}^{N,Esto} + \sum_q D_{n,q,t}^{N,Nsto} + \sum_r v_{r,u,t}^{N,\text{dom}} + \sum_g v_{g,u,t}^{N,\text{agr}} + \sum_o v_{o,u,t}^{N,\text{ind}} + \sum_i B_{u,i,t}^{N,\text{des}} + \sum_x G_{u,x,t}^{N,\text{rel}} + b_{u,t}^{N,\text{rej}}, \quad \forall u \in U, \forall t \in T \quad (14)$$

The rejected brine can be calculated by multiplying the total seawater by a factor (β), which represents the ratio of brine to seawater flows:

$$b_{n,t}^{E, \text{rej}} = bSW_{u,t}^{\text{in}, N}, \quad \forall u \in U, \forall t \in T \quad (15)$$

$$b_{u,t}^{N, \text{rej}} = \beta SW_{u,t}^{\text{in}, N}, \quad \forall u \in U, \forall t \in T \quad (16)$$

Model for Power Production. The total produced power ($TEnergy_t$) is the sum of the energy generated in existing ($E_{\text{production}}^E$) and new ($E_{\text{production}}^N$) power desalination plants:

$$TEnergy_t = \sum_u E_{\text{production}}^N_{u,t} + \sum_n E_{\text{production}}^E_{n,t}, \quad \forall t \in T \quad (17)$$

This value is subject to the electricity demands in industrial zones ($E_{o,t}^{\text{ind}}$), domestic users ($E_{r,t}^{\text{dom}}$), and agricultural regions ($E_{r,t}^{\text{agr}}$):

$$TEnergy_t = \sum_r E_{r,t}^{\text{dom}} + \sum_g E_{g,t}^{\text{agr}} + \sum_o E_{o,t}^{\text{ind}}, \quad \forall t \in T \quad (18)$$

The energy produced in new power desalination plants is a function of their capacity. This function is represented through multiplying the capacity of the new power desalination plant ($SW_{u,t}^{\text{in}, N}$) times a factor (GEP) that represents the energy usage per unit flow rate of incoming seawater:

$$E_{\text{production}}^N_{u,t} = SW_{u,t}^{\text{in}, N} \cdot \text{GEP}, \quad \forall u \in U, \forall t \in T \quad (19)$$

Existence of New Storage Tanks. The existence, location and size of new storage tanks in each region is modeled through binary variables (y_q^{sto}), if the binary variable is equal to one, then the tank is needed, if the binary variable is equal to zero, the tank is not needed, the existence is subject to the maximum capacity of the tank ($\Theta_q^{\text{sto}, \text{max}}$) and the minimum capacity of the tank ($\Theta_q^{\text{sto}, \text{min}}$):

$$y_q^{\text{sto}} \cdot \Theta_q^{\text{sto}, \text{min}} \leq S_q^{\text{sto}} \leq y_q^{\text{sto}} \cdot \Theta_q^{\text{sto}, \text{max}} \quad (20)$$

The associated installation cost ($\text{InstCost}_{q,t}^{\text{sto}}$) is a function of a fixed cost of the tank (Z_1), the unit variable cost (Z_2), and a factor used to annualize the inversion (k_F). In addition, S_q^{max} represents the maximum capacity for storage tanks:

$$\text{InstCost}_{q,t}^{\text{sto}} = k_F [Z_1 y_q^{\text{sto}} + Z_2 (S_q^{\text{max}})^{\alpha}], \quad \forall q \in Q \quad (21)$$

Where S_q^{max} is greater than the existing water in storage tank ($S_{q,t}^N$) in any time period t .

$$S_{q,t}^N \leq S_q^{\text{max}}, \quad q \in Q, t \in T \quad (22)$$

The storage cost can be calculated by the next equation:

$$\text{StorageCost} = H_Y \sum_q \text{InstCost}_{q,t}^{\text{sto}} \quad (23)$$

Existence of New Power Desalination Plants. The existence, location, and capacity of the new power desalination plants are modeled through binary variables (y_u^{pdes}). If the binary variable is equal to one, then the power desalination plant is needed; if the binary variable is equal to zero, the power desalination plant is not needed; also, the power desalination plant has a maximum capacity ($\Theta_u^{\text{pdes}, \text{max}}$) and a minimum capacity ($\Theta_u^{\text{pdes}, \text{min}}$):

$$y_u^{\text{pdes}} \cdot \Theta_u^{\text{pdes}, \text{min}} \leq SW_u^{\text{max}} \leq y_u^{\text{pdes}} \cdot \Theta_u^{\text{pdes}, \text{max}} \quad (24)$$

The installation cost ($\text{InstCost}_{u,t}^{\text{pdes}}$) is a function of the unit costs (Z_3 and Z_4), the maximum seawater capacity (SW_u^{max}), and a factor used to annualize the inversion (k_F). The operating cost ($\text{OpCost}_{u,t}^{\text{pdes}}$) is a function of a unit operating cost (Z_5), the total water extracted from the sea ($SW_{u,t}^{\text{in}, N}$), the recovery factor ($1 - \beta$), and a factor used to account for the operational time per year (H_Y):

$$\text{InstCost}_{u,t}^{\text{pdes}} = k_F [Z_3 y_u^{\text{pdes}} + Z_4 (SW_u^{\text{max}})^{\alpha}], \quad \forall u \in U \quad (25)$$

$$\text{OpCost}_{u,t}^{\text{pdes}} = H_Y [Z_5 (1 - \beta) SW_{u,t}^{\text{in}, N}], \quad \forall u \in U, \forall t \in T \quad (26)$$

Where, SW_u^{max} is greater than any possible water in a power-desalination plant ($SW_{q,t}^N$) during any time period t .

$$SW_{u,t}^{\text{in}, N} \leq SW_u^{\text{max}}, \quad u \in U, t \in T \quad (27)$$

The operating cost is a function of the total water extracted from the sea ($SW_{n,t}^{\text{in}, E}$) multiplied by the total recovery ($1 - \beta$) and the operating unit cost (Z_6):

$$\text{OpCost}_{n,t}^{E, \text{des}} = H_Y [Z_6 (1 - \beta) SW_{n,t}^{\text{in}, E}], \quad \forall n \in N, \forall t \in T \quad (28)$$

H_Y is a factor used to account for the operational time per year.

The installation cost for new power desalination plants can be calculated as follows:

$$\text{NPDistcost} = \sum_u \text{InstCost}_u^{\text{pdes}} \quad (29)$$

The operating cost for new power desalination plants can be calculated using the next expression:

$$\text{NPDoCost} = \sum_u \sum_t \text{OpCost}_{u,t}^{\text{pdes}} \quad (30)$$

On the other hand, the total operating cost for existing power desalination plants can be calculated using the following expression:

$$\text{TEPDopcost} = \sum_n \sum_t \text{TOpCost}_{n,t}^{E, \text{des}} \quad (31)$$

It should be noticed that the total operating cost for the existing power desalination plant ($\text{TOpCost}_{n,t}^{E, \text{des}}$) also includes the cost of the fuel consumption.

The total energy requirement ($\text{TER}_{u,t}^N$) of the new power desalination plants is a function of the total seawater fed. This function has a linear behavior and depends on the capacity of the new power desalination plants ($SW_{u,t}^{\text{in}, N}$) multiplied by a factor (FCF):

$$\text{TER}_{u,t}^N = SW_{u,t}^{\text{in}, N} \cdot \text{FCF}, \quad \forall u \in U, \forall t \in T \quad (32)$$

where $\text{TER}_{u,t}^N$ is equal to the energy obtained by the combustion of fossil fuels ($Q_{f,u,t}^{\text{fossil}}$), biofuels ($Q_{b,u,t}^{\text{biofuel}}$), and the energy obtained by the solar collector ($Q_{u,t}^{\text{solar}}$) during any period t .

$$\text{TER}_{u,t}^N = \sum_{f \in F} Q_{f,u,t}^{\text{fossil}} + \sum_{b \in B} Q_{b,u,t}^{\text{biofuel}} + Q_{u,t}^{\text{solar}}, \quad \forall u \in U, \forall t \in T \quad (33)$$

The total energy cost (TEC^N) is obtained by the next equation, where FFC_f and BFC_b are the prices of fossil fuels and biofuels, respectively.

$$NPDEnergycost = H_Y \sum_u \sum_t \left[\sum_{f \in F} (FFC_f \cdot Q_{f,u,t}^{\text{fossil}}) + \sum_{b \in B} (BFC_b \cdot Q_{b,u,t}^{\text{biofuel}}) \right] \quad (34)$$

The amount of heat to be produced is subject to the availability of biofuels ($AVF_{f,t}^{\text{max}}$) and fossil fuels ($AVB_{b,t}^{\text{max}}$) multiplied by its corresponding heating power factor, HPF_f for fossil fuels and HPB_b for biofuels, according to the next relationships.

For fossil fuels

$$\sum_u Q_{f,u,t}^{\text{fossil}} \leq HPF_f \cdot AVF_{f,t}^{\text{max}}, \quad \forall f \in F, \forall t \in T \quad (35)$$

For biofuels

$$\sum_u Q_{b,u,t}^{\text{biofuel}} \leq HPB_b \cdot AVB_{b,t}^{\text{max}}, \quad \forall b \in B, \forall t \in T \quad (36)$$

Existence of Solar Collector. The existence, area, and cost of the solar collector are modeled using binary variables (w_u^{solar}). If the binary variable is equal to one, then the solar collector is needed, if the binary variable is equal to zero, the solar collector is not needed, this is activated according to the maximum ($ATot_u^{\text{max}}$) and minimum collecting area ($ATot_u^{\text{min}}$):

$$w_u^{\text{solar}} \cdot ATot_u^{\text{min}} \leq AREA_u^{\text{solar,max}} \leq w_u^{\text{solar}} \cdot ATot_u^{\text{max}} \quad (37)$$

The capital cost for the solar collector ($SCCost_u^{\text{solar}}$) is a function of the unit costs (Z_1^{solar} and Z_2^{solar}), the maximum effective solar collector area ($AREA_u^{\text{solar,max}}$), an area factor (γ), and a factor to annualize the inversion (k_F). The operating cost ($OpCost_{u,t}^{\text{solar}}$) is a function of a unit operating factor (Z_3^{solar}), the area of the solar collector in each period ($A_{u,t}^{\text{solar}}$), and a factor to account for the operational time per year (H_Y):

$$SCCost_u^{\text{solar}} = k_F [Z_1^{\text{solar}} \cdot w_u^{\text{solar}} + Z_2^{\text{solar}} (AREA_u^{\text{solar,max}})^{\alpha}], \quad \forall u \in U \quad (38)$$

$$SOCost_{u,t}^{\text{solar}} = H_Y [Z_3^{\text{solar}} \cdot A_{u,t}^{\text{solar}}], \quad \forall u \in U, \forall t \in T \quad (39)$$

where $AREA_u^{\text{solar,max}}$ is greater than the existing area of the solar collector ($A_{u,t}^{\text{solar}}$) in any time period t .

$$A_{u,t}^{\text{solar}} \leq AREA_u^{\text{solar,max}}, \quad \forall u \in U, t \in T \quad (40)$$

The heating provided by the solar collector ($Q_{u,t}^{\text{solar}}$) is obtained multiplying the useful collected energy ($UCE_{u,t}^{\text{solar}}$) by the effective area of the solar collector ($A_{u,t}^{\text{solar}}$):

$$Q_{u,t}^{\text{solar}} = UCE_{u,t}^{\text{solar}} \cdot A_{u,t}^{\text{solar}} \quad (41)$$

The total cost for the solar collector can be calculated using the next relationship:

$$\text{SolarCost} = \sum_u SCCost_u^{\text{solar}} + \sum_u \sum_t SOCost_{u,t}^{\text{solar}} \quad (42)$$

Pumping and Piping Costs. The water distribution can be calculated using the following equation:

$$\text{PipingCost} = k_F \left[\begin{aligned} & \sum_r DPC1_r \cdot \gamma_r^{h,\text{dom}} + \sum_p \sum_r DPC2_{p,r} \cdot \gamma_{p,r}^{sE,\text{dom}} + \sum_q \sum_r DPC3_{q,r} \cdot \gamma_{q,r}^{sN,\text{dom}} + \sum_g APC1_g \cdot \gamma_g^{h,\text{agr}} \\ & + \sum_p \sum_g APC2_{p,g} \cdot \gamma_{p,g}^{sE,\text{agr}} + \sum_q \sum_g APC3_{q,g} \cdot \gamma_{q,g}^{sN,\text{agr}} + \sum_o IPC1_o \cdot \gamma_o^{h,\text{ind}} + \sum_p \sum_o IPC2_{p,o} \cdot \gamma_{p,o}^{sE,\text{ind}} \\ & + \sum_q \sum_o IPC3_{q,o} \cdot \gamma_{q,o}^{sN,\text{ind}} + \sum_r \sum_j DPC4_{r,j} \cdot \gamma_{r,j}^{d,\text{dom}} + \sum_g \sum_j APC4_{g,j} \cdot \gamma_{g,j}^{d,\text{agr}} + \sum_o \sum_j IPC4_{o,j} \cdot \gamma_{o,j}^{d,\text{ind}} \\ & + \sum_r \sum_x DPC5_{r,x} \cdot \gamma_{r,x}^{w,\text{dom}} + \sum_g \sum_x APC5_{g,x} \cdot \gamma_{g,x}^{w,\text{agr}} + \sum_o \sum_x IPC5_{o,x} \cdot \gamma_{o,x}^{w,\text{ind}} + \sum_n \sum_i BPC1_{n,i} \cdot \gamma_{n,i}^{BE,\text{des}} \\ & + \sum_u \sum_i BPC2_{u,i} \cdot \gamma_{u,i}^{BN,\text{des}} + \sum_r \sum_n DPC6_{r,n} \cdot \gamma_{r,n}^{vE,\text{dom}} + \sum_r \sum_u DPC7_{r,u} \cdot \gamma_{r,u}^{vN,\text{dom}} + \sum_g \sum_n APC6_{g,n} \cdot \gamma_{g,n}^{vE,\text{agr}} \\ & + \sum_u \sum_g APC7_{g,u} \cdot \gamma_{g,u}^{vN,\text{agr}} + \sum_o \sum_n IPC6_{o,n} \cdot \gamma_{o,n}^{vE,\text{ind}} + \sum_o \sum_u IPC7_{o,u} \cdot \gamma_{o,u}^{vN,\text{ind}} + \sum_n \sum_q EPC1_{n,q} \cdot \gamma_{n,q}^{DE,\text{Nsto}} \\ & + \sum_g \sum_p EPC2_{n,p} \cdot \gamma_{n,p}^{DE,\text{Esto}} + \sum_u \sum_q NPC1_{u,q} \cdot \gamma_{u,q}^{DN,\text{Nsto}} + \sum_u \sum_p NPC2_{u,p} \cdot \gamma_{u,p}^{DN,\text{Esto}} + \sum_n \sum_x GPC1_{n,x} \cdot \gamma_{n,x}^{GE,\text{rel}} \\ & + \sum_u \sum_x GPC2_{u,x} \cdot \gamma_{u,x}^{GN,\text{rel}} + \sum_i \sum_j AQP1_{i,j} \cdot \gamma_{i,j}^{a,\text{dw}} + \sum_p \sum_i SPC1_{p,i} \cdot \gamma_{p,i}^{sE,\text{aq}} + \sum_q \sum_i SPC2_{q,i} \cdot \gamma_{q,i}^{sN,\text{aq}} \end{aligned} \right] \quad (43)$$

The piping cost factors are determined by the next equation:

$$\text{PipingCostFactor} = k_m L D^m \quad (44)$$

where L is the pipe length, D^m is the pipe diameter, k_m and m are pipe cost parameters that depend on the pipe material. Because the diameter of the pipes, the distance from the storage points to final users and height are fixed, then the flow rate determines the

piping cost. The existence of the pipe is determined by the binary variable and it is used to activate the pipe cost accounting for the maximum capacity used over all time periods. Maximum values to determine piping cost are described in the [Supporting Information](#).

The pumping cost can be calculated using the next expression:

$$\begin{aligned}
\text{PumpingCost} = H_Y \left[\begin{aligned}
& \sum_r \sum_t h_{r,t}^{\text{dom}} \text{PPD1}_{r,t} + \sum_p \sum_r \sum_t s_{p,r,t}^{E,\text{dom}} \text{PPD2}_{p,r,t} + \sum_q \sum_r \sum_t s_{q,r,t}^{N,\text{dom}} \text{PPD3}_{q,r,t} + \sum_g \sum_t h_{g,t}^{\text{agr}} \text{PPA1}_{g,t} \\
& + \sum_g \sum_p \sum_t s_{g,p,t}^{E,\text{agr}} \text{PPA2}_{g,p,t} + \sum_q \sum_g \sum_t s_{q,g,t}^{N,\text{agr}} \text{PPA3}_{q,g,t} + \sum_o \sum_t h_{o,t}^{\text{ind}} \text{PPI1}_{o,t} + \sum_o \sum_p \sum_t s_{o,p,t}^{E,\text{ind}} \text{PPI2}_{o,p,t} \\
& + \sum_g \sum_p \sum_t s_{g,o,t}^{N,\text{ind}} \text{PPI3}_{g,o,t} + \sum_q \sum_g \sum_t d_{r,j,t}^{\text{dom}} \text{PPD4}_{r,j,t} + \sum_o \sum_t d_{g,j,t}^{\text{agr}} \text{PPA4}_{g,j,t} + \sum_o \sum_p \sum_t d_{o,j,t}^{\text{ind}} \text{PPI4}_{o,j,t} \\
& + \sum_q \sum_o \sum_t w_{r,x,t}^{\text{dom}} \text{PPD5}_{r,x,t} + \sum_r \sum_j \sum_t w_{g,x,t}^{\text{agr}} \text{PPA5}_{g,x,t} + \sum_g \sum_j \sum_t w_{o,x,t}^{\text{ind}} \text{PPI5}_{o,x,t} \\
& + \sum_r \sum_x \sum_t B_{n,i,t}^{E,\text{des}} \text{PPC1}_{n,i,t} + \sum_g \sum_x \sum_t B_{u,i,t}^{N,\text{des}} \text{PPC2}_{u,i,t} + \sum_o \sum_x \sum_t v_{r,n,t}^{E,\text{dom}} \text{PPD6}_{r,n,t} \\
& + \sum_n \sum_i \sum_t v_{r,u,t}^{N,\text{dom}} \text{PPD7}_{r,u,t} + \sum_u \sum_i \sum_t v_{g,n,t}^{E,\text{agr}} \text{PPA6}_{g,n,t} + \sum_r \sum_n \sum_t v_{g,u,t}^{N,\text{agr}} \text{PPA7}_{g,u,t} \\
& + \sum_o \sum_n \sum_t v_{o,n,t}^{E,\text{ind}} \text{PPI6}_{o,n,t} + \sum_o \sum_u \sum_t v_{o,u,t}^{N,\text{ind}} \text{PPI7}_{o,u,t} + \sum_n \sum_q \sum_t D_{n,q,t}^{E,\text{Nsto}} \text{PPE1}_{n,q,t} \\
& + \sum_n \sum_p \sum_t D_{n,p,t}^{E,\text{Esto}} \text{PPE2}_{n,p,t} + \sum_u \sum_q \sum_t D_{u,q,t}^{N,\text{Nsto}} \text{PPN1}_{u,q,t} + \sum_u \sum_p \sum_t D_{u,p,t}^{N,\text{Esto}} \text{PPN2}_{u,p,t} \\
& + \sum_n \sum_p \sum_t G_{n,x,t}^{E,\text{rel}} \text{PPG1}_{n,x,t} + \sum_u \sum_q \sum_t G_{u,x,t}^{N,\text{rel}} \text{PPG2}_{u,x,t} + \sum_u \sum_p \sum_t a_{i,j,t}^{\text{dw}} \text{PAQ1}_{i,j,t} \\
& + \sum_p \sum_i \sum_t s_{p,i,t}^{E,\text{aq}} \text{PPS1}_{p,i,t} + \sum_q \sum_i \sum_t s_{q,i,t}^{N,\text{aq}} \text{PPS2}_{q,i,t}
\end{aligned} \right] \quad (45)
\end{aligned}$$

The pumping cost factors are determined by the Darcy–Weisbach equation:³⁸

$$\begin{aligned}
\text{PumpingCostFactor} \\
= \frac{1}{0.0000576} \frac{L}{D^5} \frac{(\text{no. of hours})(\$/\text{kWh})}{\eta} \quad (46)
\end{aligned}$$

where f is the friction factor, L is the pipe length, D is the pipe inner diameter, and η is the combined pump and motor efficiency. The friction factor is based on the pipe roughness, pipe diameter, and the Reynolds number.

Tax Credit Reduction. It is needed an equation to represent the revenues obtained by the reduction of GHGE as tax credits, which is obtained taking into account the reduction of GHGE by replacing fossil fuels (i.e., natural gas and oil) with biofuels and solar collectors:

$$\text{TCR} = H_Y[(\text{TF}_{\text{CO}_2}^{\text{REF}} - F_{\text{CO}_2}) \cdot R_{\text{tax}}] \quad (47)$$

$\text{TF}_{\text{CO}_2}^{\text{REF}}$ is the amount of total GHGE when only fossil fuels are used in the dual purpose power plants, and F_{CO_2} is the amount of GHGE when fossil fuels, biofuels, or solar technology is used. R_{tax} is the tax credit for CO_2 emissions.

The amount of GHGE are calculated as follows:

$$\begin{aligned}
F_{\text{CO}_2} = \sum_f \sum_u \sum_t [\text{GHGE}_f^{\text{fossil}} \cdot Q_{f,u,t}^{\text{fossil}}] \\
+ \sum_b \sum_u \sum_t [\text{GHGE}_b^{\text{biofuel}} \cdot Q_{b,u,t}^{\text{biofuel}}] \quad (48)
\end{aligned}$$

Objective Function. The proposed multiobjective optimization model involves three important aspects. The first one consists in maximizing the gross annual profit, as economic objective. The second one is the minimization of the overall GHGE, as environmental objective. Finally, the third one is the quantification of the jobs generated by the project, as a social objective.

$$\text{OF} = \text{Max AnnualProfit}; \text{Min OGHGE}; \text{Quantifying ONJobs} \quad (49)$$

The economic objective function consists in maximizing the gross annual profit, the function includes water sales, energy sales and tax credit reduction, minus the total annual cost.

$$\text{AnnualProfit} = \text{WaterSales} + \text{EnergySales} + \text{TCR} - \text{TAC} \quad (50)$$

Where the annual water sales include the water sold to domestic, industrial, and agricultural users, and it can be calculated through the next relationship:

$$\begin{aligned}
\text{WaterSales} = \\
H_Y \sum_t \left[\begin{aligned}
& (\sum_r h_{r,t}^{\text{dom}} + \sum_p \sum_r s_{p,r,t}^{E,\text{dom}} + \sum_q \sum_r s_{q,r,t}^{N,\text{dom}}) \text{WDC}_t \\
& + (\sum_g h_{g,t}^{\text{agr}} + \sum_g \sum_p s_{g,p,t}^{E,\text{agr}} + \sum_q \sum_g s_{q,g,t}^{N,\text{agr}}) \text{WAC}_t \\
& + (\sum_o h_{o,t}^{\text{ind}} + \sum_o \sum_p s_{o,p,t}^{E,\text{ind}} + \sum_q \sum_o s_{q,o,t}^{N,\text{ind}}) \text{WIC}_t
\end{aligned} \right] \quad (51)
\end{aligned}$$

The energy sales can be calculated using the next equation:

$$\begin{aligned}
\text{EnergySales} = H_Y \sum_t \left[\sum_r E_{r,t}^{\text{dom}} \cdot \text{DEC}_t + \sum_g E_{g,t}^{\text{agr}} \cdot \text{AEC}_t \right. \\
\left. + \sum_o E_{o,t}^{\text{ind}} \cdot \text{IEC}_t \right] \quad (52)
\end{aligned}$$

The total annual cost (TAC) includes the installation (NPDinstcost), operating (NPDopcost), and energy consumption cost (NPDenergycost) for new power-desalination plants, as well as the total operating costs for existing power desalination plants (TEPDopcost), the equation also includes the storage cost (StorageCost), piping cost (PipingCost), pumping cost (PumpingCost), and solar collector cost (SolarCost).

$$\begin{aligned}
\text{TAC} = \text{NPDinstcost} + \text{NPDopcost} + \text{NPDenergycost} \\
+ \text{EPDopcost} + \text{StorageCost} + \text{PipingCost} \\
+ \text{PumpingCost} + \text{SolarCost} \quad (53)
\end{aligned}$$

The environmental objective function (OGHGE) seeks to minimize the overall greenhouse gas emissions, as an indirect

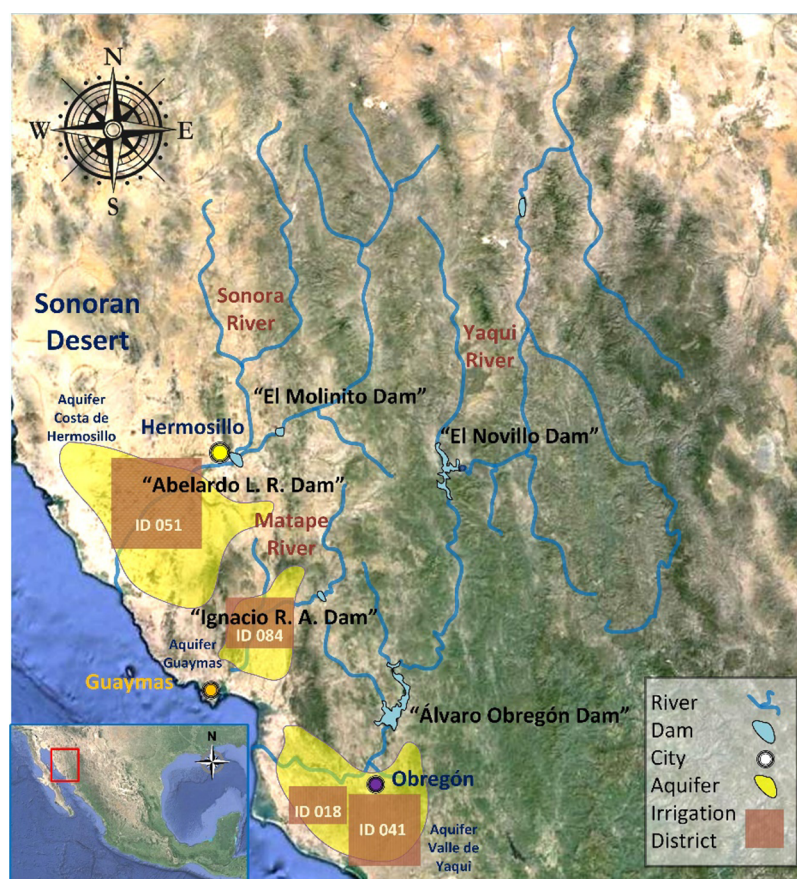


Figure 2. Overview of the addressed case study. Created using Google Earth.

environmental impact assessment. The equation takes into account the GHGE for fossil fuels ($\text{GHGE}_f^{\text{fossil}}$) and biofuels ($\text{GHGE}_b^{\text{biofuel}}$), the emissions for the solar collector is assumed to be zero, according to the next equation:

$$\begin{aligned} \min \text{ OGHGE} = & \sum_f \sum_u \sum_t [\text{GHGE}_f^{\text{fossil}} \cdot Q_{f,u,t}^{\text{fossil}}] \\ & + \sum_b \sum_u \sum_t [\text{GHGE}_b^{\text{biofuel}} \cdot Q_{b,u,t}^{\text{biofuel}}] \end{aligned} \quad (54)$$

The quantification of jobs (ONJobs) is determined indirectly through the amount of energy used (see the work of Lira-Barragan et al.³⁹) based on the JEDI model (jobs and economic development impact), in which the number of jobs can be obtained per kilowatt hour produced by fossil fuels ($\text{NJOB}_f^{\text{fossil}}$), biofuels ($\text{NJOB}_b^{\text{biofuel}}$), and solar energy ($\text{NJOB}^{\text{solar}}$); this objective can be obtained as follows:

$$\begin{aligned} \text{ONJobs} = & \sum_f \sum_u \sum_t [\text{NJOB}_f^{\text{fossil}} \cdot Q_{f,u,t}^{\text{fossil}}] \\ & + \sum_b \sum_u \sum_t [\text{NJOB}_b^{\text{biofuel}} \cdot Q_{b,u,t}^{\text{biofuel}}] \\ & + \sum_u \sum_t [\text{NJOB}^{\text{solar}} \cdot Q_{u,t}^{\text{solar}}] \end{aligned} \quad (55)$$

CASE STUDY

As a case study is addressed the water scarcity problem in Hermosillo Sonora, Mexico (Figure 2). This problem has been

described by González-Bravo et al.³⁷ The city of Hermosillo is located in the Sonoran Desert region where the low rainfall and the increasing demand have led to shortage of water from lakes, rivers and aquifers. The most critical situation occurred during 2012, when the Sonoran State Government proposed the construction of an aqueduct that would bring water from El Novillo dam to the inhabitants of the Hermosillo city. Nonetheless, there was insufficient supply from EL Novillo dam because of the water used for agricultural and drinking needs of the inhabitants of the Valle de Yaqui region. This caused conflicts between the people of Hermosillo city and the Valle del Yaqui tribe. The government decided to rule in favor of the inhabitants of the Valle del Yaqui region. Consequently, this caused a serious water shortage in the Hermosillo region.

Figure 2 shows a map of the region to be analyzed in the case study. Water demands for domestic and industrial users from Hermosillo city, Obregon city, and Guaymas city are taken into account. The agricultural users involve the irrigation districts 018, 041, 051, and 084. The considered water bodies include the Sonoran River, Yaqui River, and Matape River; also the volume of water content in the “El Novillo” dam, “Alvaro Obregon” dam, “Ignacio R. A.” dam, “Abelardo L. R. dam”, and “El Molinito dam” is considered. The study includes the aquifers of the region, Costa de Hermosillo, “Guaymas”, and “Valle de Yaqui”. It should be noted that the Abelardo L. R. and El Molinito dams are drained and the Costa de Hermosillo, Guaymas, and Valle de Yaqui aquifers are overexploited. Additional information has been presented by Gonzalez-Bravo et al.³⁷ The Hermosillo region is near to the Sonoran desert, and this represents an attractive location for solar collectors, due to high direct normal

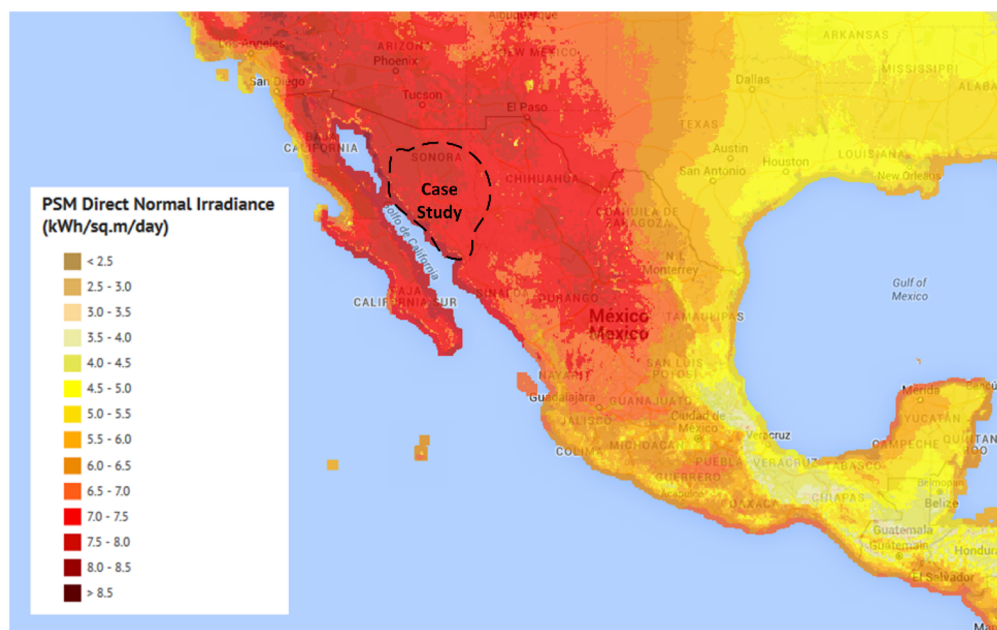


Figure 3. DNI irradiation for the case study (Reprinted from ref 40. Copyright 2012 NREL).

irradiation (DNI) values (around 3000 kWh/m²) as can be seen in Figure 3.⁴⁰ Figure 4 shows the total energy received by the

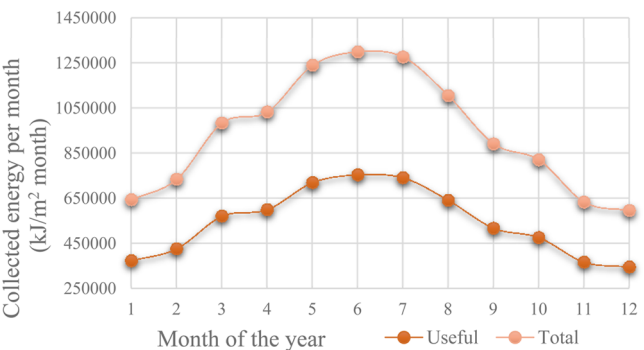


Figure 4. Useful collected energy for the case study. Data taken from ref 40.

solar collector and the useful energy in each month based on a solar collector efficiency of 50%.⁴¹ It should be noted, in Figure 4, that the highest DNI is always around noon, reaching its extreme between May and August. The solar collector was modeled according to the data reported by Liqreina and Qoaidar.²³ For this case, the dual purpose power plant was modeled using the data reported by a similar facility operated by the Qatar Electricity and Water Company (QEWC).⁴² The QEWC is the

second largest company producing power and water desalination in the Middle East and North Africa (MENA region). This company generates electricity by 5432 MW and produces 258 million gallons of water per day. The fuels and biofuels taken into account are presented in Table 1, which contains their heating power, overall GHGE, unit cost, and number of jobs created by each fuel, whereas Table 2 shows the maximum availability in each month for biofuels, this availability depends on the seasonality as well as the production of agricultural wastes.^{38,43} For the specific case of Mexico, there are not restrictions for the availability of fossil fuels to operate the dual purpose power plants.

RESULTS AND DISCUSSION

The proposed model was coded in the software GAMS. The solvers DICOPT in conjunction with CONOPT and CPLEX were used to solve this model.⁴⁴ The model consists of 9714 equations, 10 058 continuous variables, and 720 binary variables. The model considers three main objectives: Profit, OGHGE, and NJOBS. The model was solved using the constraint method to obtain the Pareto curve taking into account only two objectives (Profit vs OGHGE) because these objectives are reconciled. Meanwhile ONJOBS are evaluated for each point on the Pareto curve.

The results are presented through a Pareto curve (see Figure 6), where the points A, B, C, D, and E can be identified. Point A

Table 1. Data for Fossil Fuels and Biofuels^{38,43}

fuel	cost (\$/MM kJ)	heating power (kJ/kg)	overall GHGE (ton CO ₂ equiv/kJ)	number of jobs (jobs/kJ)
Fossil				
oil	26.195	45200	8.05408×10^{-8}	1.81677×10^{-11}
natural gas	2.554	54000	7.90892×10^{-8}	5.25431×10^{-11}
Biofuels				
biomass	1.980	17200	2.44307×10^{-8}	6.6964×10^{-8}
biogas	7.215	52,000	2.68216×10^{-8}	5.25431×10^{-7}
biodiesel	30.920	40,200	5.13283×10^{-8}	2.46582×10^{-6}
bioethanol	13.915	29,600	5.8436×10^{-8}	2.87453×10^{-6}

Table 2. Maximum Availability of Fuels and Biofuels (kg/month)^{38,43}

fuel/month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
biomass	10000	10000	40000	50000	70000	150000	100000	50000	40000	40000	30000	20000
biogas	5000	5000	6000	6500	6500	10000	10000	10000	8000	7000	6000	5000
biodiesel	5000	5500	6000	7000	10000	10000	10000	10000	9000	8000	7000	6000
bioethanol	10000	11000	12000	12000	15000	15000	15000	15000	14000	13000	11000	10000

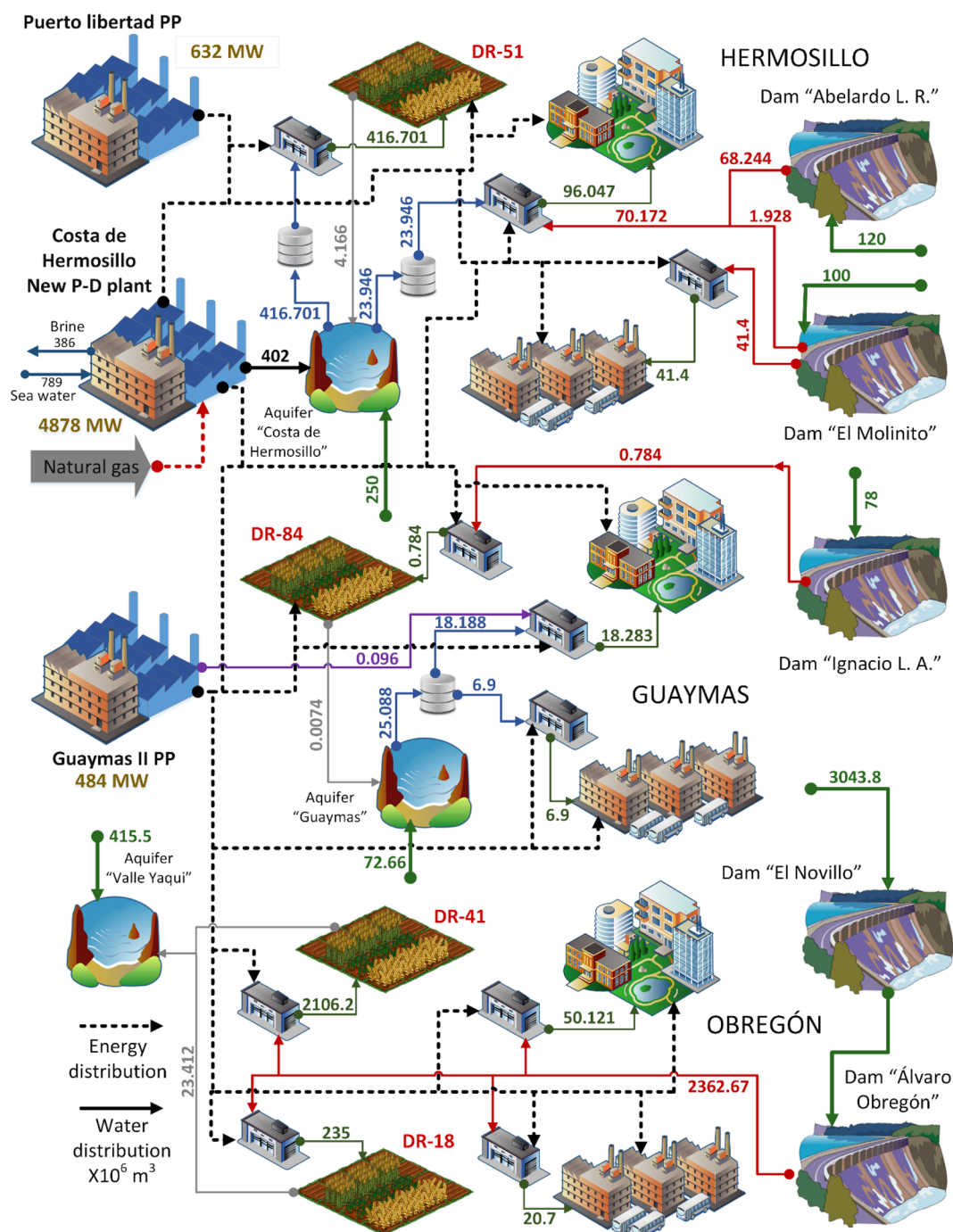


Figure 5. Optimal solution for point A.

represents the optimal results maximizing the Profit, in this case a total of $1.45 \text{ ton CO}_2 \times 10^7/\text{y}$ are released, and a total of 10 307 jobs are generated. In this case, the optimal configuration for water and electricity distribution is presented in Figure 5. As it was expected, the results are similar to the results presented in

González-Bravo et al.,³⁷ where the dual purpose power plant is installed in the Hermosillo beach at time period 1, and the total amount of water produced in the dual purpose power plant is injected to the Costa de Hermosillo aquifer ($402 \times 10^6 \text{ m}^3/\text{y}$), the aquifer also receives water from agricultural users (4.2×10^6

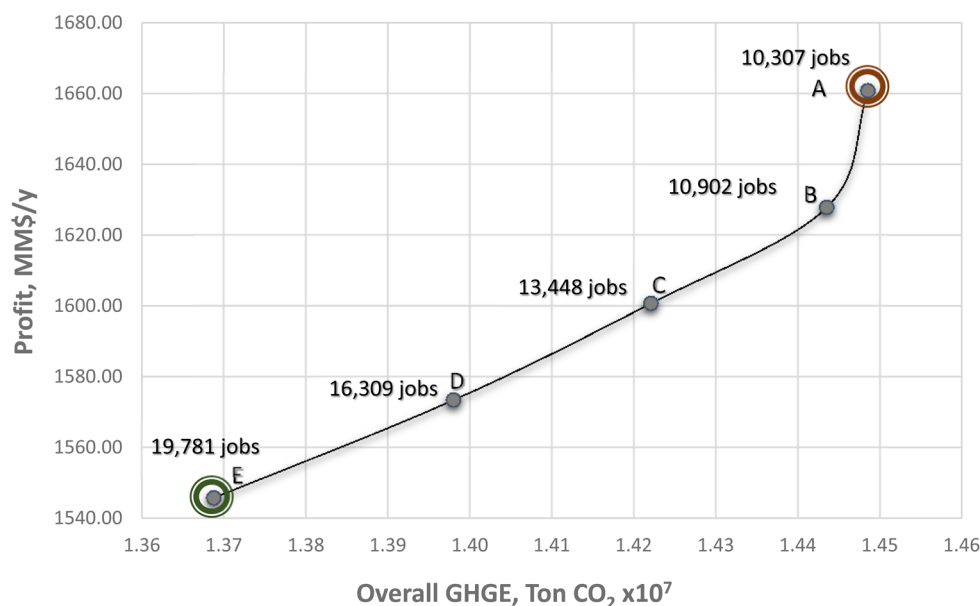


Figure 6. Pareto curve for the case study.

m^3/y) and from natural recharge ($250 \times 10^6 \text{ m}^3/\text{y}$). The agricultural users extract $416.7 \times 10^6 \text{ m}^3/\text{y}$ of water from the aquifer. The water for domestic users is taken from the Costa de Hermosillo aquifer ($23.95 \times 10^6 \text{ m}^3/\text{y}$), El Molinito dam ($1.93 \times 10^6 \text{ m}^3/\text{y}$), and from the Abelardo L. R. dam ($68.2 \times 10^6 \text{ m}^3/\text{y}$). The water for industrial users is taken from the El Molinito dam ($41.4 \times 10^6 \text{ m}^3/\text{y}$). In the case of the Guaymas region, the Guaymas aquifer receives water from natural recharge ($78 \times 10^6 \text{ m}^3/\text{y}$) and for agricultural users ($7.4 \times 10^3 \text{ m}^3/\text{y}$). The agricultural users extract water from the Ignacio L. A. dam ($0.784 \times 10^6 \text{ m}^3/\text{y}$). The domestic demands are supplied with water from the Guaymas aquifer ($18.2 \times 10^6 \text{ m}^3/\text{y}$) and from water produced in the Guaymas II power plant ($9.6 \times 10^4 \text{ m}^3/\text{y}$). While industrial demands are supplied with water from the aquifer ($6.9 \times 10^6 \text{ m}^3/\text{y}$). The agricultural users from the irrigation district 041 ($2106.2 \times 10^4 \text{ m}^3/\text{y}$) and irrigation district 018 ($235 \times 10^4 \text{ m}^3/\text{y}$) in Obregon are supplied by water from the Alvaro Obregon dam. Also, domestic users ($50.12 \times 10^4 \text{ m}^3/\text{y}$) and industrial users ($20.7 \times 10^4 \text{ m}^3/\text{y}$) are supplied with water from the Alvaro Obregon and El Novillo dams. Points B, C, D, and E have similar results in the water distribution network. In point A, the new dual purpose power desalination plant has a maximum capacity of 4880 MW and $1.4 \times 10^6 \text{ m}^3$ per day. In this case, the sales of water reach 1,552 MM\$/y, and the sales of energy are 1,346.3 MM\$/y. The total annual cost of the plant is 1237.4 MM\$/y, which accounts for the cost of fuels (467.4 MM\$/y), the operational cost of a new power desalination plant (592 MM\$/y), piping (1.1 MM\$/y) and pumping costs (65.7 MM\$/y), and other operational costs of the existing power plants (80 MM\$/y).

The economic results are shown in Figure 6. Point A has an annual net profit of 1,660.9 MM\$/y, for this point the CO_2 emissions are $1.45 \times 10^7 \text{ ton CO}_2 \text{ equiv/y}$ and 10 307 jobs are generated. Point B has an annual net profit of 1,627.9 MM\$/y, annual CO_2 emissions of $1.44 \times 10^7 \text{ ton CO}_2 \text{ eq/y}$, and 10 902 jobs are generated. Point C has a total annual net profit of 1,600.8 MM\$/y, point C also has $1.42 \times 10^7 \text{ ton CO}_2 \text{ equiv/y}$, and 13 449 jobs are generated. Point D has an annual profit of 1,573.5 MM\$/y, $1.40 \times 10^7 \text{ ton CO}_2 \text{ equiv/y}$ of CO_2 emissions, and 16 309 jobs are generated. Point E has an annual net profit of

1,545.9 MM\$/y, it represents 6% less than the optimal solution of point A; this reduction is mainly due to the fixed cost of the solar collector, which has virtually the same annualized cost as the installation cost of the dual purpose power plant. In point E, there is a reduction of CO_2 emissions of $7.98 \times 10^5 \text{ ton CO}_2 \text{ equiv/y}$ due to the installation of the solar collector. In this case, 19 781 jobs are generated. The capacity of the dual purpose power plant for points B–E has the same capacity as point A (4880 MW); however, there are small differences in the fuel consumption. Figure 7 shows the annual fuel consumption for

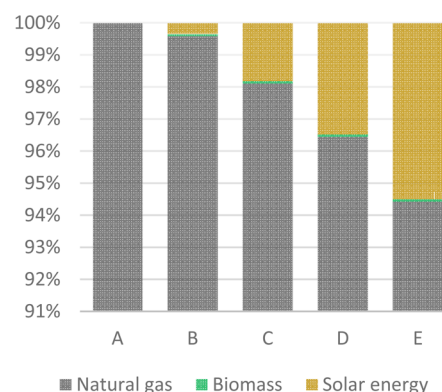


Figure 7. Fuel consumption for each Pareto point.

each point on the Pareto curve. The total energy requirement in point A is totally fulfilled using natural gas, in this scenario no renewable energy (biofuels or solar energy) was consumed. On the other hand, points B–E also include energy from both biomass and solar. However, the use of biofuels is still very limited for this region due to limited availability of biomass while the use of solar energy is restricted in this case by the capital cost and land requirements.

In terms of acceptability, the construction of this project would satisfy energy and water requirements of 784 342 inhabitants and 117 360 ha of crops in the Hermosillo region, the water and energy requirements of 409.310 inhabitants and 255.672 ha of crops in the Obregon region, and 149.299 inhabitants and 20.042

ha of crops in the Guaymas region. The project would also boost the development of 200 relevant companies throughout the region.⁴⁵ It is important to analyze the effect of the tax credit benefit due to the reduction of GHGE associated with the use of clean energies. In this case, tax credits of \$5, \$10, \$20, and \$30/ton of CO₂ equiv were evaluated. Table 3 shows the effect over

Table 3. Sensitivity Analysis for Different Values of the Tax Credit

scenario	total annual profit				
	without tax credit reduction	\$5/ton of CO ₂ equiv	\$10/ton of CO ₂ equiv	\$20/ton of CO ₂ equiv	\$30/ton of CO ₂ equiv
A	1660.9	1660.9	1660.9	1660.9	1660.9
B	1627.9	1628.4	1628.9	1629.9	1631.4
C	1600.8	1602.3	1603.8	1606.8	1611.3
D	1573.5	1576	1578.5	1583.5	1591
E	1545.9	1549.9	1553.9	1561.9	1573.9

the total annual profit for the tax credit reduction. In general, the total annual profit increases when the tax credit reduction increases; however, it is significant only when the unit tax credits are over \$20/ton of CO₂ equiv and the overall GHGE are greater than 5×10^5 ton of CO₂ equiv.

CONCLUSIONS

This paper has presented a multiobjective mathematical programming model for the optimal design of water distribution networks involving dual purpose power plants and renewable energy. The considered objectives are the maximization of the net annual profit and the minimization of the overall greenhouse gas emissions while tracking the overall number of new jobs. The optimal water distribution network and the existence of new dual purpose power plants is determined using binary variables, and the energy requirements are satisfied using fossil fuels, biofuels and solar energy. The optimal solution is subject to the water and electricity demands of domestic, agricultural and industrial users. A systematic solution approach for this multiobjective optimization model has been presented. The approach involves the solution of the multiobjective optimization formulation and the construction of a Pareto curve to show the trade-offs of the involved objectives.

The water scarcity problem in the Sonoran Desert in the northwest of Mexico has been considered as a case study. The results show important economic benefits due to the installation of the new dual purpose power plant. Additionally, the integration of renewable energy (biofuels and solar energy) can be an attractive option to satisfy water and energy in water stressed areas with high solar radiations; however, availability and cost of biofuels as well as restrictions for the needed land for solar technology still represent disadvantages to compete against fossil fuels. The trade-offs have been presented and analyzed for the various solution scenarios. Future works must take into account fluctuations for prices and availability of fossil fuels through a stochastic optimization formulation.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acssuschemeng.6b01817.

Restrictions used to calculate piping cost as well as the description of the used nomenclature. The existence of the pipe is determined by the binary variable, and it is used to activate the pipe cost accounting for the maximum capacity flow rate over all time periods. Additional data used to perform the simulation as well as their corresponding references (PDF)

AUTHOR INFORMATION

Corresponding Author

*E-mail: jmponce@umich.mx. Tel.: +52 443 3223500 x 1277. Fax.: +52 443 3273584 (J.M.P.-O.).

Notes

The authors declare no competing financial interest.

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NOMENCLATURE

Subscripts

- b* biofuels
- f* fossil fuels
- g* location of agricultural users
- i* existing aquifer
- j* deep wells
- n* location of existing power desalination plants
- o* location of industrial users
- p* location of existing water storage tanks
- q* possible location of new water storage tanks
- r* location of domestic users
- t* distribution time in months
- u* possible location of new power—desalination plants
- x* existing dam as natural resources

Positive Variable

- $a_{i,j,t}^{dw}$ water sent to the deep wells from aquifers, m³ × 10⁶/y
- AnnualProfit annual profit from water sales, \$ × 10⁶
- $AREA_u^{solar,max}$ maximum effective solar collector area, m²
- $A_{u,t}^{solar}$ effective solar collector area, m²
- $b_{n,t}^{rej}$ water sent to the sea as reject from existing power desalination plants, m³ × 10⁶/y
- $l_{u,t}^{N,rej}$ water sent to the sea as reject from new power desalination plants, m³ × 10⁶/y
- $B_{n,i,t}^{E,des}$ water received from existing power desalination plants, m³ × 10⁶/y
- $B_{u,i,t}^{N,des}$ water received from new power desalination plants, m³ × 10⁶/y
- $d_{r,j,t}^{dom}$ water distributed in central stations from deep wells (domestic), m³ × 10⁶/y
- $d_{g,j,t}^{agr}$ water distributed in central stations from deep wells (agricultural), m³ × 10⁶/y
- $d_{o,j,t}^{ind}$ water distributed in central stations from deep wells (industrial), m³ × 10⁶/y
- $D_{n,p,t}^{E,Esto}$ water received in storage tanks from existing power desalination plants, m³ × 10⁶/y
- $D_{u,p,t}^{N,Esto}$ water received in storage tanks from new power desalination plants, m³ × 10⁶/y
- $E_{r,t}^{dom}$ energy demand by domestic users, kW × 10⁶
- $E_{g,t}^{agr}$ energy demand by agricultural users, kW × 10⁶
- $E_{o,t}^{ind}$ energy demand by industrial users, kW × 10⁶

EnergySales	energy sales from power desalination plants, \$ × 10 ⁶	SW _{n,t} ^{in,E}	total water extracted from the sea for existing power desalination plants, m ³ × 10 ⁶ /y
Eproduction _{n,t} ^E	energy produced in existing power desalination plants, kW × 10 ⁶	SW _{u,t} ^{in,N}	total water extracted from the sea for new power desalination plants, m ³ × 10 ⁶ /y
TEPDopcost	total operating cost for existing power desalination plants, \$ × 10 ⁶	TAC	total annual cost, \$ × 10 ⁶
Eproduction _{u,t} ^N	energy produced in new power desalination plants, kW × 10 ⁶	TCR	tax credit reduction, \$ × 10 ⁶
F _{i,t} ^{agr}	aquifer recharge from agricultural users, m ³ × 10 ⁶ /y	TEC ^N	total energy cost, \$ × 10 ⁶
F _{CO₂}	amount of GHGE emitted in the dual purpose power plant, ton of CO ₂ equiv	TEnergy _t	total energy produced, kW × 10 ⁶
G _{n,x,t} ^{E,rel}	water received in dams from existing power desalination plants, m ³ × 10 ⁶ /y	TER _{u,t} ^{E,dom}	total energy requirement, kJ × 10 ⁶
G _{u,x,t} ^{N,rel}	water received in dams from new power desalination plants, m ³ × 10 ⁶ /y	v _{r,n,t} ^{E,dom}	water received in central stations from existing power desalination plants (domestic), m ³ × 10 ⁶ /y
h _{r,t} ^{dom}	water in domestic central station, m ³ × 10 ⁶ /y	v _{g,n,t} ^{E,agr}	water received in central stations from existing power-desalination plants (agricultural), m ³ × 10 ⁶ /y
h _{g,t} ^{agr}	water in agricultural central station, m ³ × 10 ⁶ /y	v _{o,n,t} ^{E,ind}	water received in central stations from existing power desalination plants (industrial), m ³ × 10 ⁶ /y
h _{o,t} ^{ind}	water in industrial central station, m ³ × 10 ⁶ /y	v _{r,u,t} ^{N,dom}	water received in central stations from new power desalination plants (domestic), m ³ × 10 ⁶ /y
H _Y	operating hours for the plant per year, h/y	v _{g,u,t} ^{N,agr}	water received in central stations from new power desalination plants (agricultural), m ³ × 10 ⁶ /y
InstCost _q ^{sto}	storage installation cost, \$ × 10 ⁶	v _{o,u,t} ^{N,ind}	water received in central stations from new power desalination plants (domestic), m ³ × 10 ⁶ /y
InstCost _u ^{pdes}	installation cost (new power desalination plants), \$ × 10 ⁶	WaterSales	water sales from power desalination plants, \$ × 10 ⁶
k _F	factor used to annualize the inversion, 1/y	W _{i,t} ^{aq}	water in aquifer, m ³ × 10 ⁶ /y
M _{x,t}	water in dam, m ³ × 10 ⁶ /y	W _{r,x,t} ^{dom}	water received in central stations from dams (domestic), m ³ × 10 ⁶ /y
NPDinstcost	installation cost for new power desalination plants, \$ × 10 ⁶	W _{g,x,t} ^{agr}	water received in central stations from dams (agricultural), m ³ × 10 ⁶ /y
NPDopcost	operational cost for new power desalination plants, \$ × 10 ⁶	W _{o,x,t} ^{ind}	water received in central stations from dams (industrial), m ³ × 10 ⁶ /y
OpCost _{u,t} ^{pdes}	operational cost (new power desalination plants), \$ × 10 ⁶	Parameters	
TOPCost _{n,t} ^{E,des}	total operating cost (existing power desalination plants), \$ × 10 ⁶	AEC _t	agricultural energy cost, \$ × 10 ⁶ /kW × 10 ⁶
pca	percentage of water that seeps into the ground	APC	agricultural piping cost, \$ × 10 ⁶ /m ³ × 10 ⁶
PipingCost	piping cost, \$ × 10 ⁶	AQP	deep well piping cost, \$ × 10 ⁶ / × 10 ⁶ m ³
PumpingCost	pumping cost, \$ × 10 ⁶	agr _{dem} _{g,t}	water requirements of agricultural users, m ³ /y
Q	heat, kJ × 10 ⁶ /y	ATot ^{min}	minimum available area of solar collector, m ²
R _{i,t} ^{aq}	natural recharge of water, m ³ × 10 ⁶ /y	ATot ^{min}	maximum available area of solar collector, m ²
SolarCost	total solar collector cost, \$ × 10 ⁶	AVF ^{max}	maximum amount available for fossil fuels, kg × 10 ⁶ /month
SCC _u ^{solar}	capital cost of solar collector, \$ × 10 ⁶	AVB ^{max}	maximum amount available for biofuels, kg × 10 ⁶ /month
SOC _{u,t} ^{solar}	operating cost of solar collector, \$ × 10 ⁶	BPC	aquifer recharge piping cost, \$ × 10 ⁶ /m ³ × 10 ⁶
S _{p,i,t} ^{E,aq}	water received from existing storage tanks, m ³ × 10 ⁶ /y	BFC	biofuel cost, \$ × 10 ⁶ /kJ × 10 ⁶
S _{q,i,t} ^{N,aq}	water received from new storage tanks, m ³ × 10 ⁶ /y	DEC _t	unit domestic energy cost, \$ × 10 ⁶ /kW × 10 ⁶
S _{p,r,t} ^{E,dom}	water received from existing storage tanks (domestic), m ³ × 10 ⁶ /y	DPC	unit domestic piping cost, \$ × 10 ⁶ /m ³ × 10 ⁶
S _{q,r,t} ^{N,dom}	water received from new storage tanks (domestic), m ³ × 10 ⁶ /y	dom _{dem} _{r,t}	water requirements of domestic users, m ³ /y
S _{p,i,t} ^{E,agr}	water received from existing storage tanks (agricultural), m ³ × 10 ⁶ /y	EPC	storage recharge piping cost (existing PD plant), \$ × 10 ⁶ /m ³ × 10 ⁶
S _{g,q,t} ^{N,agr}	water received from new storage tanks (agricultural), m ³ × 10 ⁶ /y	FCF	fuel consumption factor, kJ × 10 ⁶ /m ³ × 10 ⁶
S _{o,q,t} ^{E,ind}	water received from existing storage tanks (industrial), m ³ × 10 ⁶ /y	FFC	fossil fuel cost, \$ × 10 ⁶ /kJ × 10 ⁶
S _{o,q,t} ^{N,ind}	water received from new storage tanks (industrial), m ³ × 10 ⁶ /y	GEP	energy generation factor, kW × 10 ⁶ /m ³ × 10 ⁶
S _{p,t} ^E	water in existing storage tanks, m ³ × 10 ⁶ /y	GPC	unit dam recharge piping cost, \$ × 10 ⁶ /m ³ × 10 ⁶
S _{q,t} ^N	water in new storage tanks, m ³ × 10 ⁶ /y	HPF _f	heating power for fossil fuels, kJ × 10 ⁶ /kg × 10 ⁶
S _q ^{max}	maximum storage value, m ³ × 10 ⁶	HPB _b	heating power for biofuels, kJ × 10 ⁶ /kg × 10 ⁶
StorageCost	storage cost, \$ × 10 ⁶	IEC _t	unit industrial energy cost, \$ × 10 ⁶ /kW × 10 ⁶
SW _u ^{max}	maximum seawater consumption, m ³ × 10 ⁶ /y	IPC	unit industrial piping cost, \$ × 10 ⁶ /m ³ × 10 ⁶
		ind _{dem} _{o,t}	water requirements of industrial users, m ³ /y

NPC	unit storage recharge piping cost (new power desalination plant), $\$ \times 10^6/\text{m}^3 \times 10^6$
PPA	unit agricultural pumping cost, $\$ \times 10^6/\text{m}^3 \times 10^6$
PAQ	unit deep well pumping cost, $\$ \times 10^6/\text{m}^3 \times 10^6$
PPC	unit aquifer recharge pumping cost, $\$ \times 10^6/\text{m}^3 \times 10^6$
PPD	unit domestic pumping cost, $\$ \times 10^6/\text{m}^3 \times 10^6$
PPE	unit storage recharge piping cost (existing power desalination plant), $\$ \times 10^6/\text{m}^3 \times 10^6$
PPG	unit dam recharge pumping cost, $\$ \times 10^6/\text{m}^3 \times 10^6$
PPI	unit industrial pumping cost, $\$ \times 10^6/\text{m}^3 \times 10^6$
PPN	unit storage recharge piping cost (new power desalination plants), $\$ \times 10^6/\text{m}^3 \times 10^6$
PPS	unit storage pumping cost, $\$ \times 10^6/\text{m}^3 \times 10^6$
R_{tax}	tax credit for CO_2 , $\$ \times 10^6/\text{ton of } \text{CO}_2 \text{ equiv}$
SPC	unit storage piping cost, $\$ \times 10^6/\text{m}^3 \times 10^6$
$\text{SW}_{\text{u,t}}^{\text{N,max}}$	maximum value of seawater extracted from the sea, $\text{m}^3 \times 10^6/\text{y}$
$\text{TF}_{\text{CO}_2}^{\text{REF}}$	total GHGE when only fossil fuels are used, ton of $\text{CO}_2 \text{ equiv}$
$\text{UCE}_{\text{u,t}}^{\text{solar}}$	useful collected energy, $\text{kJ} \times 10^6/\text{m}^2 \text{ month}$
WAC_t	unit cost of water for agricultural use, $\$ \times 10^6/\text{m}^3 \times 10^6$
WDC_t	unit cost of water for domestic use, $\$ \times 10^6/\text{m}^3 \times 10^6$
WIC_t	unit cost of water for industrial use, $\$ \times 10^6/\text{m}^3 \times 10^6$
Z_1	unit fixed cost of the tank, $\$ \times 10^6$
Z_2	unit fixed cost of the tank, $\$ \times 10^6$
Z_3	unit fixed cost of new power desalination plants, $\$ \times 10^6$
Z_4	unit fixed cost of new power desalination plants, $\$ \times 10^6$
Z_5	unit fixed cost of new power desalination plants, $\$ \times 10^6$
Z_6	unit fixed cost of new power desalination plants, $\$ \times 10^6$
Z_{solar}	unitary cost of solar collector, $\$ \times 10^6$
α	economic scale factor
β	reject factor of power desalination plants
$\Theta_{\text{q}}^{\text{sto,max}}$	maximum capacity of the tank, $\text{m}^3 \times 10^6$
$\Theta_{\text{q}}^{\text{sto,min}}$	minimum capacity of the tank, $\text{m}^3 \times 10^6$
$\Theta_{\text{u}}^{\text{pdes,max}}$	maximum capacity of new power desalination plants, $\text{m}^3 \times 10^6/\text{y}$
$\Theta_{\text{u}}^{\text{pdes,min}}$	minimum capacity of the new power desalination plants, $\text{m}^3 \times 10^6/\text{y}$

Binary Variables

$w_{\text{u}}^{\text{solar}}$	existence of a solar collector
$y_{\text{q}}^{\text{sto}}$	existence of storage tanks
$y_{\text{u}}^{\text{pdes}}$	existence of power–desalination plants

REFERENCES

- (1) WWAP, *The United Nations World Water Development Report 2015: Water and Energy*; United Nations World Water Assessment Programme, UNESCO, Paris, France, 2015.
- (2) Davis, R. J.; Kim, Y.; Logan, B. E. Increasing desalination by mitigating anolyte pH imbalance using catholyte effluent addition in a multi-anode bench scale microbial desalination cell. *ACS Sustainable Chem. Eng.* **2013**, *1*, 1200–1206.
- (3) IRENA. *Water Desalination Using Renewable Energy: Technology Brief*; International Renewable Energy Agency, Abu Dhabi, United Arab Emirates, 2012.
- (4) Almansoori, A.; Saif, Y. Structural optimization of osmosis processes for water and power production in desalination applications. *Desalination* **2014**, *344*, 12–27.

- (5) Esfahani, I. J.; Kim, J. T.; Yoo, C. K. A Cost approach for optimization of a combined power and thermal desalination system through exergy and environmental analysis. *Ind. Eng. Chem. Res.* **2013**, *52* (32), 11099–11110.
- (6) Zhang, S.; Zhang, Y.; Chung, T.-S. Facile preparation of antifouling hollow fiber membranes for sustainable osmotic power generation. *ACS Sustainable Chem. Eng.* **2016**, *4*, 1154–1160.
- (7) Raluy, R. G.; Serra, L.; Uche, J.; Valero, A. Lifecycle assessment of desalination technologies integrated with energy production systems. *Desalination* **2004**, *167*, 445–458.
- (8) Shaffer, D. L.; Chavez, L. H. A.; Ben-Sasson, M.; Castrillón, S. R.-V.; Yip, N. Y.; Elimelech, M. Desalination and reuse of high-salinity shale gas produced water: drivers, technologies, and future directions. *Environ. Sci. Technol.* **2013**, *47*, 9569–9583.
- (9) Gutiérrez-Arriaga, C. G.; Serna-González, M.; Ponce-Ortega, J. M.; El-Halwagi, M. M. Sustainable integration of algal biodiesel production with steam electric power plants for greenhouse gas mitigation. *ACS Sustainable Chem. Eng.* **2014**, *2*, 1388–1403.
- (10) Tavakkoli, S.; Lokare, O. R.; Vidic, R. D.; Khanna, V. Systems-level analysis of waste heat recovery opportunities from natural gas compressor stations in the United States. *ACS Sustainable Chem. Eng.* **2016**, *4*, 3618–3626.
- (11) Sternberg, A.; Bardow, A. Life cycle assessment of power-to-gas: Syngas vs methane. *ACS Sustainable Chem. Eng.* **2016**, *4*, 4156–4165.
- (12) Miller, S.; Shemer, H.; Semiat, R. Energy and environmental issues in desalination. *Desalination* **2015**, *366*, 2–8.
- (13) Lueken, C.; Cohen, G. E.; Apt, J. Costs of solar and wind power variability for reducing CO_2 emissions. *Environ. Sci. Technol.* **2012**, *46*, 9761–9767.
- (14) Ravi, S.; Lobell, D. B.; Field, C. B. Tradeoffs and synergies between biofuel production and large solar infrastructure in deserts. *Environ. Sci. Technol.* **2014**, *48*, 3021–3030.
- (15) Börjesson, K.; Lennartson, A.; Moth-Poulsen, K. Efficiency limit of molecular solar thermal energy collecting devices. *ACS Sustainable Chem. Eng.* **2013**, *1*, 585–590.
- (16) Mabrouk, A. A.; Fath, H. E. S. Techno-economic analysis of hybrid high performance MSF desalination plant with NF membrane. *Desalin. Water Treat.* **2013**, *51* (4–6), 844–856.
- (17) Mabrouk, A. A.; Fath, H. E. S. Experimental study of high performance hybrid NF-MSF desalination pilot test unit driven by renewable energy. *Desalin. Water Treat.* **2013**, *51* (37–39), 6895–6904.
- (18) López-Díaz, D. C.; Lira-Barragán, L. F.; Rubio-Castro, E.; Ponce-Ortega, J. M.; El-Halwagi, M. M. Synthesis of eco-industrial parks interacting with a surrounding watershed. *ACS Sustainable Chem. Eng.* **2015**, *3*, 1564–1578.
- (19) Tovar-Facio, J.; Lira-Barragán, L. F.; Nápoles-Rivera, F.; Bamufleh, H. S.; Ponce-Ortega, J. M.; El-Halwagi, M. M. Optimal synthesis of refinery property-based water networks with electro-coagulation treatment systems. *ACS Sustainable Chem. Eng.* **2016**, *4*, 147–158.
- (20) Gutierrez-Arriaga, C. G.; Serna-Gonzalez, M.; Ponce-Ortega, J. M.; El-Halwagi, M. M. Multi-objective optimization of steam power plants for sustainable generation of electricity. *Clean Technol. Environ. Policy* **2013**, *15*, 551–566.
- (21) Iaquaniello, G.; Salladini, A.; Mari, A.; Mabrouk, A. A.; Fath, H. E. S. Concentrating solar power (CSP) system integrated with MED–RO hybrid desalination. *Desalination* **2014**, *336*, 121–128.
- (22) Liqueina, A.; Qoaidar, L. Dry cooling of concentrating solar power (CSP) plants, an economic competitive option for the desert regions of the MENA region. *Sol. Energy* **2014**, *103*, 417–424.
- (23) Qoaidar, L.; Liqueina, A. Optimization of dry cooled parabolic trough (CSP) plants for the desert regions of the Middle East and North Africa (MENA). *Sol. Energy* **2015**, *122*, 976–985.
- (24) Fthenakis, V.; Atia, A. A.; Morin, O.; Bkayrat, R.; Sinha, P. New prospects for PV powered water desalination plants: Case studies in Saudi Arabia. *Prog. Photovoltaics* **2016**, *24*, 543–550.
- (25) Sankar, D.; Deepa, N.; Rajagopal, S.; Karthik, K. M. Solar power and desalination plant for carbon black industry: Improved techniques. *Sol. Energy* **2015**, *119*, 243–250.

- (26) Gorjian, S.; Ghobadian, B. Solar desalination: A sustainable solution to water crisis in Iran. *Renewable Sustainable Energy Rev.* **2015**, *48*, 571–584.
- (27) Abdelhady, F.; Bamufleh, H.; El-Halwagi, M. M.; Ponce-Ortega, J. M. Optimal design and integration of solar thermal collection, storage, and dispatch with process cogeneration systems. *Chem. Eng. Sci.* **2015**, *136*, 158–167.
- (28) Palenzuela, P.; Alarcón-Padilla, D. C.; Zaragoza, G.; Blanco, J. Comparison between CSP+MED and CSP+RO in Mediterranean area and MENA region: Techno-economic analysis. *Energy Procedia* **2015**, *69*, 1938–1947.
- (29) Palenzuela, P.; Alarcón-Padilla, D. C.; Zaragoza, G. Large-scale solar desalination by combination with CSP: Techno-economic analysis of different options for the Mediterranean Sea and the Arabian Gulf. *Desalination* **2015**, *366*, 130–138.
- (30) Diaf, A.; Cherfa, A.; Karadaniz, L.; Tigrine, Z. A technical–economical study of solar desalination. *Desalination* **2016**, *377*, 123–127.
- (31) Liu, S.; Konstantopoulou, F.; Gikas, P.; Papageorgiou, L. G. A mixed integer optimization approach for integrated water resources management. *Comput. Chem. Eng.* **2011**, *35* (5), 858–875.
- (32) Atilhan, S.; Linke, P.; Abdel-Wahab, A.; El Halwagi, M. M. A systems integration approach to the design of regional water desalination and supply networks. *Int. J. Proc. Sys. Eng.* **2011**, *1* (2), 125–135.
- (33) Atilhan, S.; Mahfouz, A.; Batchelor, B.; Linke, P.; Abdel-Wahab, A.; Nápoles-Rivera, F.; Jiménez-Gutiérrez, A.; El-Halwagi, M. M. A systems-integration approach to the optimization of macroscopic water desalination and distribution networks: a general framework applied to Qatar's water resources. *Clean Technol. Environ. Policy* **2012**, *14* (2), 161–171.
- (34) Nápoles-Rivera, F.; Serna-González, M.; El-Halwagi, M. M.; Ponce-Ortega, J. M. Sustainable water management for macroscopic systems. *J. Cleaner Prod.* **2013**, *47*, 102–117.
- (35) Nápoles-Rivera, F.; Rojas-Torres, M. G.; Ponce-Ortega, J. M.; Serna-González, M.; El-Halwagi, M. M. Optimal design of macroscopic water networks under parametric uncertainty. *J. Cleaner Prod.* **2015**, *88*, 172–184.
- (36) Alnouri, S. Y.; Linke, P.; El-Halwagi, M. M. A synthesis approach for industrial city water reuse networks considering central and distributed treatment systems. *J. Cleaner Prod.* **2015**, *89*, 231–250.
- (37) González-Bravo, R.; Nápoles-Rivera, F.; Ponce-Ortega, J. M.; El-Halwagi, M. M. Involving integrated seawater desalination-power plants in the optimal design of water distribution networks. *Resour. Conserv. Recycl.* **2015**, *104*, 181–193.
- (38) Swamee, P. K.; Sharma, A. K. *Design of Water Supply Pipe Networks*; Wiley-Interscience: Hoboken, New Jersey, USA, 2008.
- (39) Lira-Barragan, L. F.; Ponce-Ortega, J. M.; Serna-González, M.; El-Halwagi, M. M. Synthesis of integrated absorption refrigeration systems involving economic and environmental objectives and quantifying social benefits. *Appl. Therm. Eng.* **2013**, *52*, 402–419.
- (40) NREL. *National Solar Radiation Database*; National Renewable National Laboratory (U.S. Department of Energy), Washington DC, USA, 2012.
- (41) Weinstein, L. A.; Loomis, J.; Bhatia, B.; Bierman, D. M.; Wang, E. N.; Chen, G. Concentrating solar power. *Chem. Rev.* **2015**, *115*, 12797–12838.
- (42) QEWC, *Qatar Electricity and Water Company Annual Report*; Qatar Electricity and Water Company, Doha, Qatar, 2014.
- (43) EIA, U.S., *Independent Statics and Analysis*; Energy Information Administration, Washington DC, USA, 2016.
- (44) Brooke, A.; Kendrick, D.; Meeruas, A.; Raman, R. *GAMS-language guide*; GAMS Development Corporation, Washington DC, USA, 2016.
- (45) INEGI. Consultation System for Census Information. National Institute of Statistics and Geography, México. 2012; <http://gaia.inegi.org.mx/scince2/viewer.html> (accessed June 2016).